Department of Computer Science & Engineering

PROJECT SYNOPSIS

Semester: 8th (Session 2023-24)

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Project Title: -

Sign Language recognition using machine learning.

Synopsis of Project:

Introduction:

The project aims to develop a real-time sign language recognition application using machine learning to interpret pre-defined American Sign Language (ASL) gestures. By detecting hand gestures in real-time and converting them into text, the application addresses communication barriers faced by individuals unable to vocalize speech, thus providing technological assistance to overcome expressive challenges.

Hardware and Software Requirements:

Hardware:

- Computer or mobile device with a camera
- Sufficient memory and processing power for real-time image processing

Software:

- Python programming language
- Machine learning libraries such as TensorFlow or PyTorch
- OpenCV for image processing
- User interface development tools (e.g., Tkinter or PyQt)

Application of Project:

The developed application allows users to:

1. Communicate through hand gestures in real-time.
2. Detect and interpret pre-defined ASL gestures accurately.
3. Store detected characters in a text file for future reference.
4. Customize gestures to suit individual communication needs.

Future Work:

1. Enhance Gesture Recognition: Improve the accuracy and robustness of gesture recognition algorithms by collecting more diverse datasets and fine-tuning machine learning models.
2. Expand Gesture Vocabulary: Incorporate a broader range of ASL gestures to increase the application's versatility and utility for users.
3. Multi-Language Support: Extend the application to recognize and interpret sign language gestures from other languages, catering to a more diverse user base.
4. Accessibility Features: Implement additional features such as voice feedback or haptic feedback to assist users with visual impairments or additional disabilities.
5. Integration with Other Technologies: Explore integration possibilities with augmented reality (AR) or virtual reality (VR) platforms to create immersive and interactive sign language learning experiences.

Conclusion:

The sign language recognition system developed in this project leverages machine learning techniques to provide real-time interpretation of ASL gestures, offering a valuable communication tool for individuals facing speech-related challenges. With further refinement and expansion, the system holds the potential to significantly enhance accessibility and inclusivity for users in various communication contexts.

Name of the Guide: Manoj Kumar Singh **Signature ……**

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